

# OpenMM MD Simulation Execution

Goal: Run equilibration and production MD simulations using OpenMM on the CHARMM-GUI output files.

## PHASE 1: Environment Preparation

### 1.1. File Preparation

Extract the CHARMM-GUI output archive and navigate into the openmm directory:

```
tar -xzf charmm-gui.tgz
cd charmm-gui-[JOB_ID]/openmm
```

### 1.2. Activate Conda Environment

```
conda activate md_run
```

Note: The md\_run environment contains OpenMM, MDAAnalysis, MDTraj and all required dependencies for both simulation and post-MD analysis.

## PHASE 2: Equilibration

A 6-step equilibration process is run to gradually relax the system, starting with strong restraints on the protein and membrane, and progressively releasing them. This prevents sudden movements that could crash the simulation.

### 2.1. Automated Equilibration (All 6 Steps)

Paste the following loop into the terminal to run steps 6.1 through 6.6 sequentially:

```
init=step5_input
equi_prefix=step6.%d_equilibration

for cnt in 1 2 3 4 5 6; do
  pcnt=$((cnt-1))
  istep=$(printf $equi_prefix $cnt)
  pstep=$(printf $equi_prefix $pcnt)
  input_param="-t toppar.str -p ${init}.psf -c ${init}.crd"

  if [ $cnt -eq 1 ]; then
    input_param="$input_param -b sysinfo.dat"
  fi

  if [ $cnt -ne 1 ]; then
    input_param="$input_param -irst ${pstep}.rst"
  fi

  echo "Starting: ${istep}..."
  python -u openmm_run.py -i ${istep}.inp ${input_param} \
    -orst ${istep}.rst -odcd ${istep}.dcd > ${istep}.out
done
```

## PHASE 3: Production

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### 3.1. Duration Adjustment

Edit the `step7_production.inp` file to set the simulation length. The `nstep` parameter controls the number of MD steps. Each step is 2 fs, so:

- 25 ns = 12,500,000 steps
- 50 ns = 25,000,000 steps
- 500 ns = 250,000,000 steps (used in this project, split across 10 runs)

Note: For 500 ns total, we run 10 separate 50 ns production runs (`step7_1` through `step7_10`) rather than one long run. This avoids data loss if the server crashes and makes trajectory analysis more manageable.

### 3.2. Run First Production Segment

```
python -u openmm_run.py \  
-i step7_production.inp \  
-t toppar.str \  
-p step5_input.psf \  
-c step5_input.crd \  
-first step6.6_equilibration.rst \  
-orst step7_1.rst \  
-odcd step7_1.dcd > step7_1.out
```

### 3.3. Continue Production (Segments 2–10)

Each subsequent segment restarts from the previous `.rst` file:

```
python -u openmm_run.py \  
-i step7_production.inp \  
-t toppar.str \  
-p step5_input.psf \  
-c step5_input.crd \  
-first step7_1.rst \  
-orst step7_2.rst \  
-odcd step7_2.dcd > step7_2.out
```

Repeat with incrementing numbers (`step7_2.rst` → `step7_3.rst`, etc.) until `step7_10.dcd` is complete.

Note: At the end of production you will have: `step7_1.dcd`, `step7_2.dcd`, ..., `step7_10.dcd`. Each file contains 1,000 frames (50 ns at 50 ps/frame), giving 10,000 frames total for 500 ns.